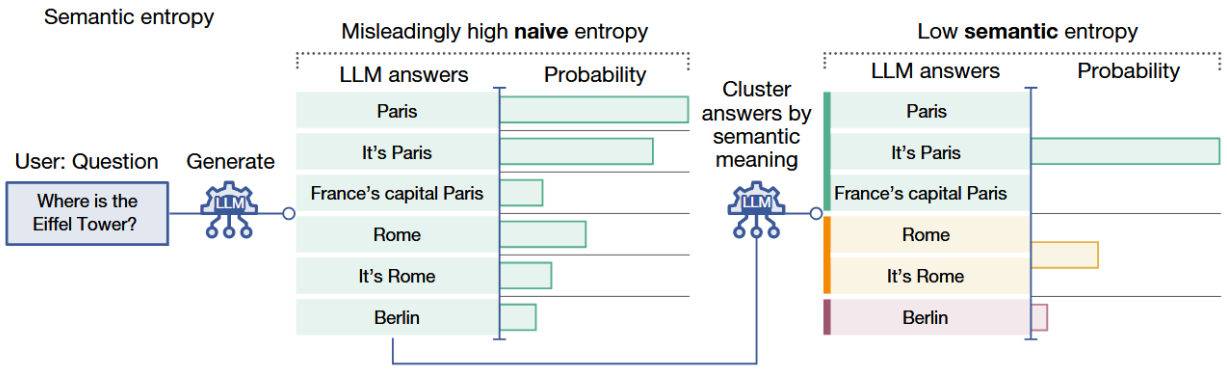
Semantic Entropy

Sematic Entropy[32]



Algorithm 1 Bidirectional Entailment Clustering

Require: context x, set of seqs. $\{\mathbf{s}^{(2)}, \ldots, \mathbf{s}^{(M)}\}$, NLI classifier $\mathcal{M}$, set of meanings $C = \{\{\mathbf{s}^{(1)}\}\}$
for $2 \leq m \leq M$ **do**
  for $c \in C$ **do** ▷ Compare to already-processed meanings.
    $\mathbf{s}^{(c)} \leftarrow c_0$ ▷ Use first sequence for each semantic-class.
    `left` $\leftarrow \mathcal{M}(\text{cat}(x, \mathbf{s}^{(c)}, \text{“<g/>”}, x, \mathbf{s}^{(m)}))$ ▷ Does old sequence entail new one?
    `right` $\leftarrow \mathcal{M}(\text{cat}(x, \mathbf{s}^{(m)}, \text{“<g/>”}, x, \mathbf{s}^{(c)}))$ ▷ Vice versa?
    if `left` is `entailment` **and** `right` is `entailment` **then**
      $c \leftarrow c \bigcup \mathbf{s}^{(m)}$ ▷ Put into existing class.
    end if
  end for
  $C \leftarrow C \bigcup \{\mathbf{s}^{(m)}\}$ ▷ Semantically distinct, gets own class.
end for
return C

Main Steps:

1. **Generation:** Sample **M** sequences {s(1), . . . , s(M)} from the predictive distribution.
2. **Clustering:** Cluster the sequences which mean the same thing. (use a Deberta-large[45] model that is fine-tuned on the NLI data set MNLI[46].)
3. **Entropy estimation:** Approximate semantic entropy by summing probabilities and compute resulting entropy.

$$SE(x) = -\sum_c p(c \mid x) \log p(c \mid x)$$

$$= -\sum_c \left(\left(\sum_{\mathbf{s} \in c} p(\mathbf{s} \mid x) \right) \log \left[\sum_{\mathbf{s} \in c} p(\mathbf{s} \mid x) \right] \right).$$

(a) Scenario 1: No semantic equivalence

Answer s	Likelihood $p(\mathbf{s} \mid x)$	Semantic likelihood $\sum_{\mathbf{s} \in c} p(\mathbf{s} \mid x)$
Paris	0.5	0.5
Rome	0.4	0.4
London	0.1	0.1
Entropy	0.94	0.94

(b) Scenario 2: Some semantic equivalence

Answer s	Likelihood $p(\mathbf{s} \mid x)$	Semantic likelihood $\sum_{\mathbf{s} \in c} p(\mathbf{s} \mid x)$
Paris	0.5	0.9
It's Paris	0.4	
London	0.1	0.1
Entropy	0.94	0.33

[32] Detecting hallucinations in large language models using semantic entropy, Nature 2024

Sematic Entropy[32]

- **Task Type:** Open-ended/Math generation

- **Introduction:** Semantic Entropy measures LLM uncertainty in semantic space, by quantifying how many distinct meanings its free-form outputs can take. It samples multiple generations, groups them into meaning-equivalent clusters using NLI-based entailment, and computes entropy over the cluster probabilities. Semantic Entropy predicts answer correctness more reliably than lexical entropy and self-evaluation.

Type	Uncertainty Sources	Model Accessibility	Training Necessity	Consumption	Evaluation
Consensus-based	Decoding/Token-level	White-box	Post-hoc	Medium	Ranking

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